Vinay Kumar Korrapati

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PROFESSIONAL SUMMARY

- Java Software Developer with 4+ years of experience applying Java 8/11, Spring Boot, and microservices to build scalable, high-performance backend services that support distributed workloads and maintain strict availability requirements.
- Java and Spring Boot components engineered to deliver reliable RESTful APIs, improving request handling throughput by 32% while sustaining predictable performance under high concurrent user demand.
- Microservices architecture patterns applied to decompose monoliths into independent services, increasing deployment velocity by 30% and improving long-term maintainability across backend platforms.
- AWS and cloud-native services incorporated to strengthen infrastructure resilience, enabling 99.9% uptime for core backend systems and reducing environment-specific failures across deployments.
- Docker and Kubernetes orchestration applied to standardize service packaging, improving runtime consistency by 28% and supporting efficient scaling during peak traffic periods.
- MySQL, PostgreSQL, and MongoDB queries optimized to reduce latency by 35% while maintaining reliable access patterns for transactional and NoSQL storage needs.
- CI/CD pipelines enhanced with Jenkins and GitHub Actions to automate builds, tests, and deployments, reducing manual overhead by 40% and ensuring consistent release quality.
- Object-oriented design patterns implemented to simplify complex backend logic, improving extensibility by 26% and reducing long-term refactoring efforts across modules.
- JVM tuning and performance optimization applied to stabilize high-traffic service endpoints, lowering response times by 33% and improving user experience under sustained load.
- Cross-functional collaboration executed with QA, DevOps, and Product teams to ensure clarity of technical requirements, reducing rework cycles and improving feature delivery accuracy.
- Code review participation leveraged to reinforce best practices and maintain coding standards, improving overall code hygiene and reducing defect introduction across the service ecosystem.
- Debugging and issue-triage workflows applied to resolve production bottlenecks quickly, improving MTTR by 38% and ensuring stable execution across distributed backend environments.

PROFESSIONAL SKILLS

- **Programming & Frameworks:**
  Java 8/11, Spring Boot, Spring Framework, REST APIs, Microservices, OOP, Design Patterns, Node.js, Python, TypeScript
- **Cloud & DevOps:**
  AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitLab CI, GitHub Actions, CI/CD Automation, Cloud-Native Services
- **Databases:**
  MySQL, PostgreSQL, MongoDB, Oracle, Cassandra, Redis, NoSQL, Relational Database Modeling
- **Security & Compliance:**
  OAuth2, JWT, IAM, Secure Coding Standards, Encryption, Audit Logging, High-Availability Architecture
- **Testing & Quality:**
  JUnit, Mockito, Integration Testing, Performance Testing, Debugging, Code Quality, Automated Testing Pipelines
- **Collaboration & Delivery:**
  Agile, Scrum, Cross-Functional Teams, Architecture Discussions, Technical Communication, Code Reviews, Peer Mentoring
- **Additional Tools & Expertise:**
  Distributed Systems, High-Performance Services, Scalability Optimization, API Design, Observability, Troubleshooting

PROFESSIONAL EXPERIENCE

Software Engineer — **02/2025 - Present**
General Motors, Detroit, Michigan

- Java 11, Spring Boot, and microservice components developed to scale financial backend platforms, improving request throughput by 32% while sustaining strict availability requirements under distributed workloads.
- REST API contracts structured with consistent domain models to reduce integration errors by 28% and ensure predictable behavior across internal service communication layers.
- AWS EC2, S3, and IAM configurations applied to enhance backend reliability, enabling 99.9% uptime and reducing infrastructure-related incidents across business-critical applications.
- Docker and Kubernetes orchestration executed to standardize deployment workflows, improving runtime consistency by 33% for high-traffic backend services.
- MySQL and Cassandra queries optimized to reduce data-access latency by 35% while maintaining secure and accurate transaction retrieval across financial systems.
- CI/CD pipelines enhanced with Jenkins and GitHub Actions to automate build and test stages, decreasing deployment friction by 40% across multi-environment releases.
- JVM performance tuning and thread optimization applied to lower average response times by 30% during peak system load for financial service endpoints.
- Cross-functional collaboration executed with QA and DevOps teams to clarify technical acceptance criteria, improving sprint efficiency and lowering rework frequency.
- Distributed system patterns implemented to improve service reliability, enabling smoother failover behavior and stronger resilience during node-level failures.
- Logging and observability enhancements integrated using CloudWatch and ELK to decrease MTTR by 38% and improve triage accuracy during service disruptions.
- Code review participation applied to enforce design standards and improve maintainability, reducing long-term refactor costs and strengthening backend consistency.
- Cache strategies and connection pooling parameters tuned to cut redundant database calls by 27% and improve high-volume transaction handling.
- Security controls integrated using IAM and encrypted data pathways to support compliance requirements across sensitive financial operations.
- Technical documentation produced for APIs, workflows, and service patterns to improve knowledge sharing and speed onboarding across backend engineering teams.

Software Engineer **02/2024 - 02/2025**

Elevance Health, Indianapolis, Indiana

- Java and Spring Boot backend features implemented to support healthcare data workflows, improving API responsiveness by 40% while ensuring compliance with protected data handling.
- Microservice boundaries defined and refined to improve modularity by 30%, enabling simpler deployment and versioning across cloud-based healthcare analytics systems.
- AWS Lambda, EC2, and S3 integrations structured to maintain 99.9% uptime for analytics services powering regulated healthcare decision flows.
- PostgreSQL, MongoDB, and BigQuery queries optimized to reduce report-generation latency by 40% across clinical analytics pipelines.
- CI/CD pipelines enhanced to introduce automated test gating, decreasing regression defects by 33% during continuous integration cycles.
- Containerized service workflows deployed via Docker to improve environment consistency across development, QA, and production environments.
- JUnit and Mockito testing expanded to increase coverage and stabilize high-risk modules that supported sensitive healthcare data exchanges.
- API documentation and schema definitions created to improve alignment between backend, data engineering, and product teams.
- Data transformation and validation logic executed to eliminate malformed payloads, improving reliability of ETL-driven healthcare workflows.
- Technical discussions and architecture refinement supported to ensure clear implementation pathways for HIPAA-sensitive backend services.
- Logging and trace instrumentation added using ELK to reduce debugging time by 45% for intermittent issues in distributed data pipelines.
- Secure authentication flows and encrypted communication patterns applied to maintain HIPAA and SOC2-compliant engineering standards.
- Agile collaboration maintained to ensure feature clarity and sprint predictability, lowering rework across multi-team deliverables.
- Mentorship contributions provided to help new developers adopt Java, Spring, and microservice best practices within regulated environments.

Software Engineer **08/2021 - 12/2023**

HSBC, Pune, India

- Java 8/11 and Spring Boot services engineered to support high-volume transaction systems, improving throughput by 40% across global banking operations.
- RESTful APIs structured with strong domain modeling to decrease integration inconsistencies by 25% across international payment channels.
- Cassandra and MySQL optimization strategies executed to reduce query latency by 35%, enabling faster financial record retrieval.
- Docker deployments standardized to improve build-to-run parity, reducing environment-related defects across enterprise release cycles.
- Kubernetes-based microservice distribution applied to improve program scalability and simplify rolling deployments for global customer workloads.
- CI/CD pipelines strengthened with multi-stage validations, lowering release failures by 70% and improving production stability.
- JVM tuning and connection pool optimization performed to reduce CPU usage and stabilize backend service performance under sustained peak conditions.
- Distributed tracing and ELK logging integrated to decrease debugging cycles by 42% while improving visibility into transactional flows.
- OAuth2 and JWT-based authentication patterns applied to support SOC2-aligned secure customer data access.
- Data consistency checks and schema validations incorporated to prevent malformed transaction inputs, reducing downstream processing anomalies.
- Technical discussions and architecture collaboration supported to ensure alignment across high-volume payment service teams.
- Agile story refinement executed to ensure testable, traceable, and clear development requirements for backend delivery.
- Code review participation applied to strengthen coding standards and improve maintainability across critical financial systems.
- Peer support and knowledge sharing delivered to promote effective microservice design and secure Java engineering practices across teams.

EDUCATION

Master's in Computer Science, PURDUE UNIVERSITY **2024 - 2025**

Coursework: Software Engineering, Advanced Algorithms Design, Cloud Computing, Software Testing, Game Design, Software Design, Human Computer Interaction, Low Code No Code.

PROJECTS

CodeCritic AI – AI Powered Code Review Platform - CodeCritic-Demo

Tech Stack: FastAPI, Python 3.8+, SQLAlchemy, PostgreSQL, OpenAI GPT-4 API, GitHub API/OAuth, Next.js 14, React, TypeScript, Tailwind CSS, Axios, JWT Authentication, Uvicorn

- Engineered an intelligent code analysis platform that automates GitHub repository reviews using OpenAI GPT models, reducing manual code review time by 80% and identifying security vulnerabilities, bugs, and quality issues with 85% accuracy.
- Built scalable full-stack architecture with FastAPI backend (Python), Next.js 14 frontend (TypeScript), PostgreSQL database, and GitHub OAuth integration, processing 50+ repositories and analyzing 100K+ lines of code across multiple programming languages.